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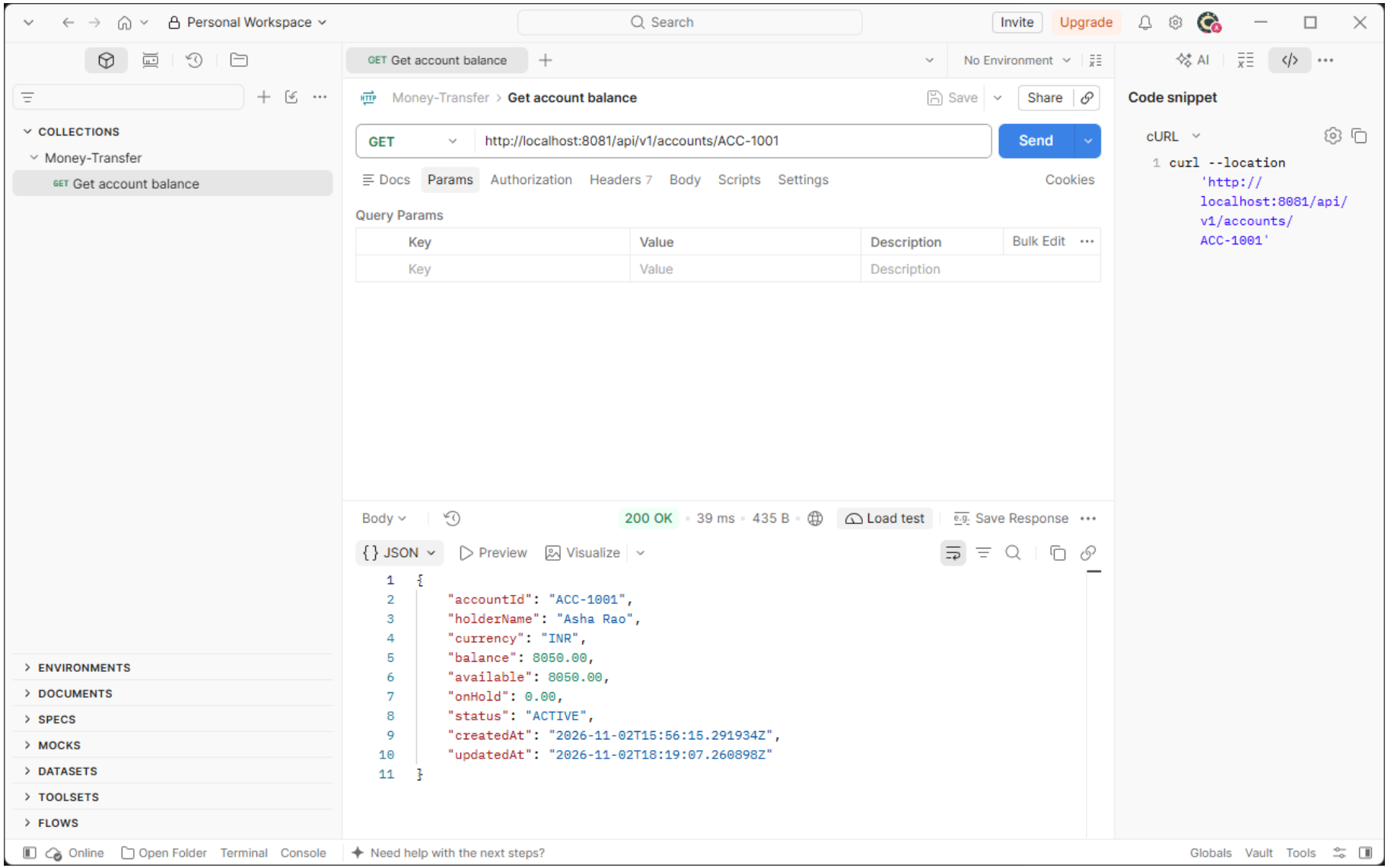
WORK INTEGRATED
LEARNING PROGRAMMES

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division

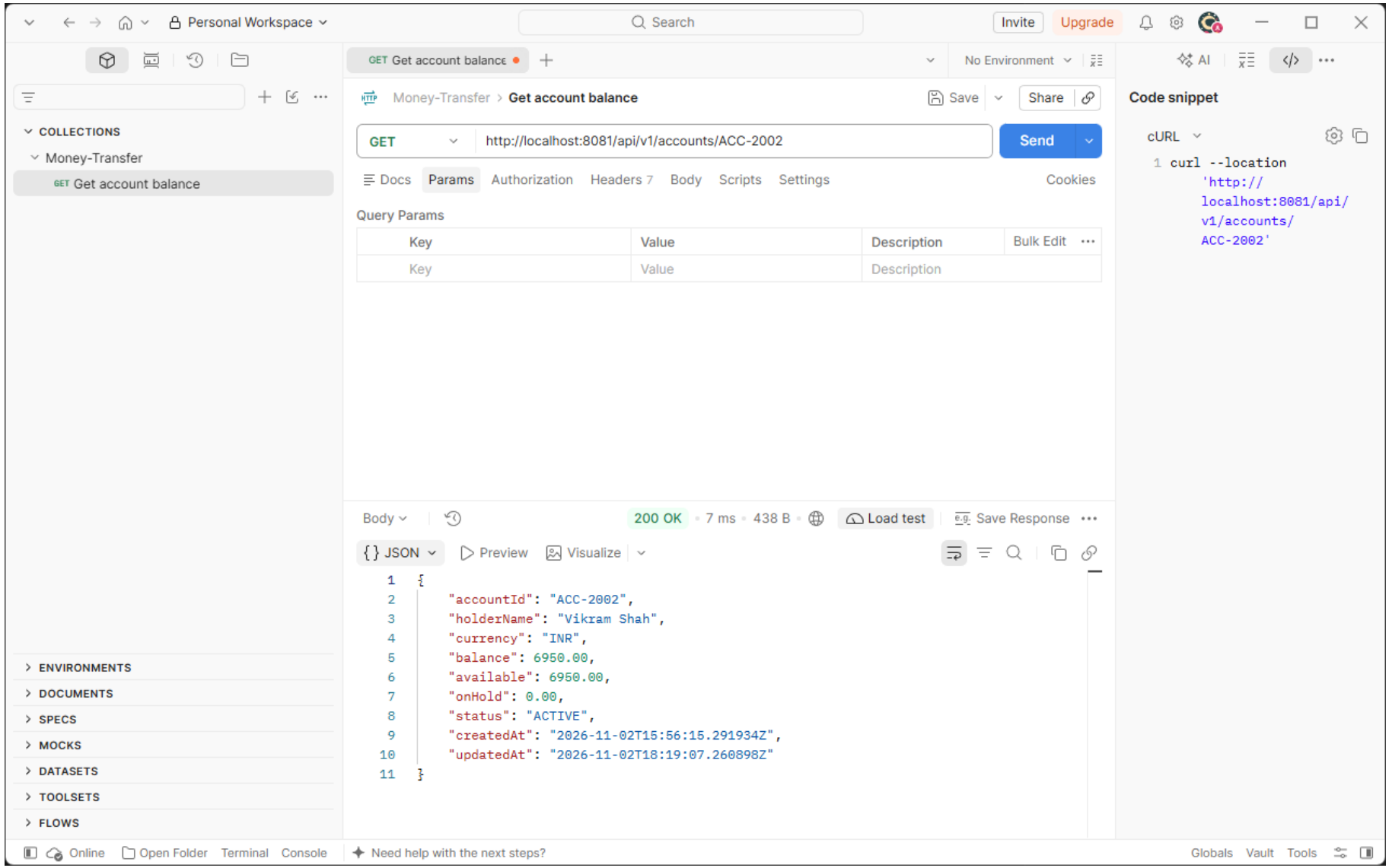
Assignment – 1 (Evidence)

SEZG583 - Scalable Services
Prof. Akanksha Bharadwaj

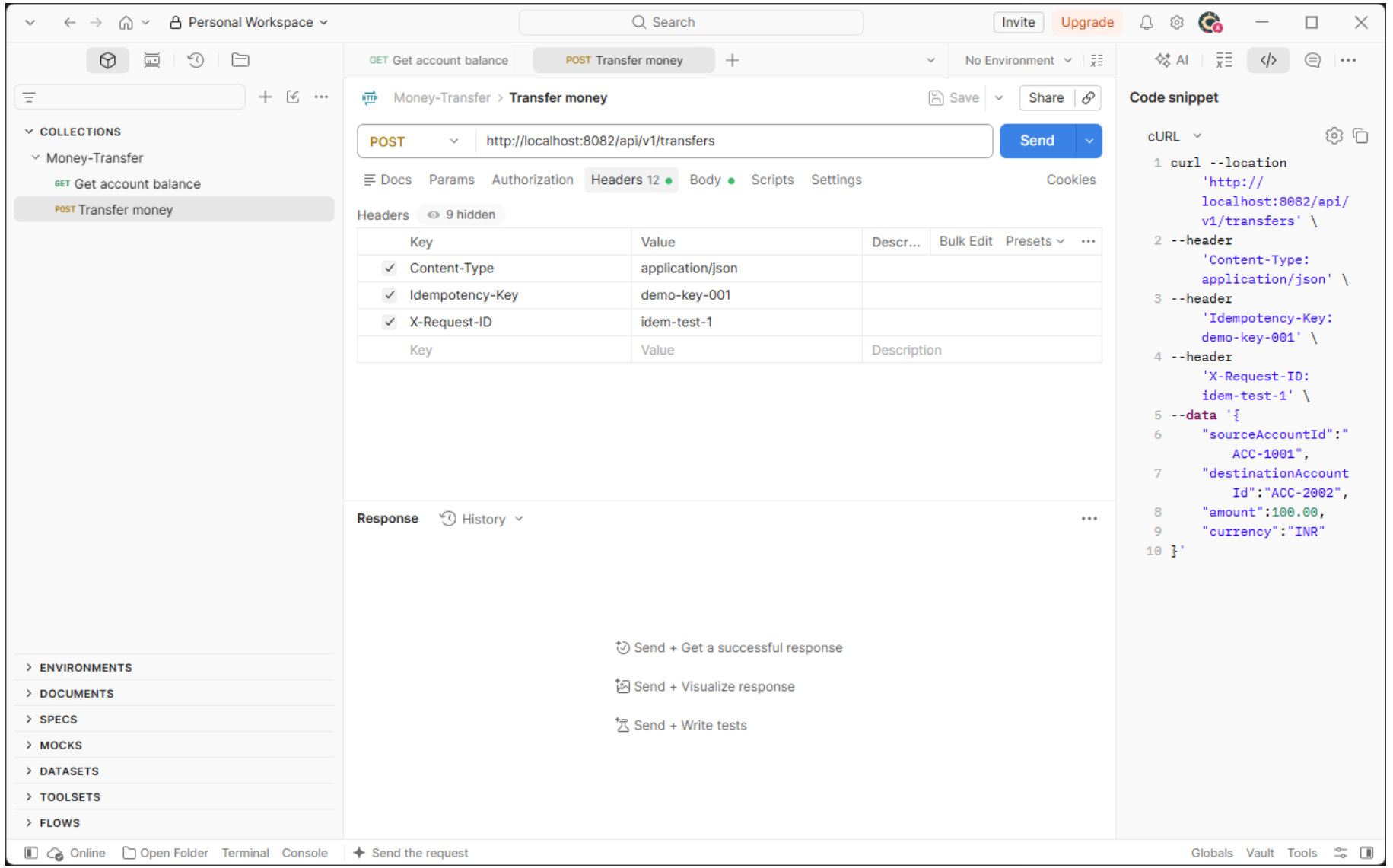
Prepared By:
Bhavesh Naphade
2025TM93185



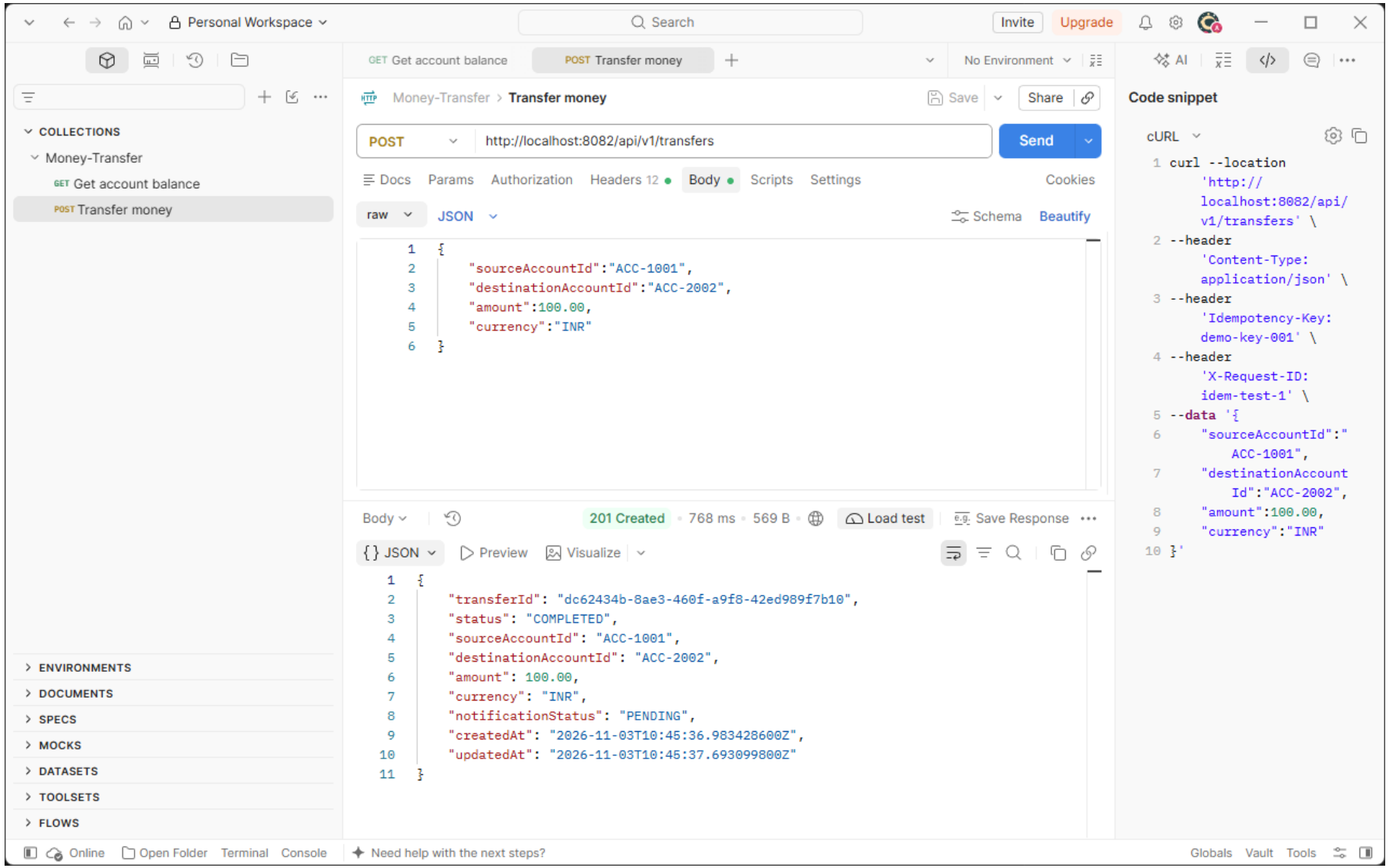
Checking account balance of account ACC-1001



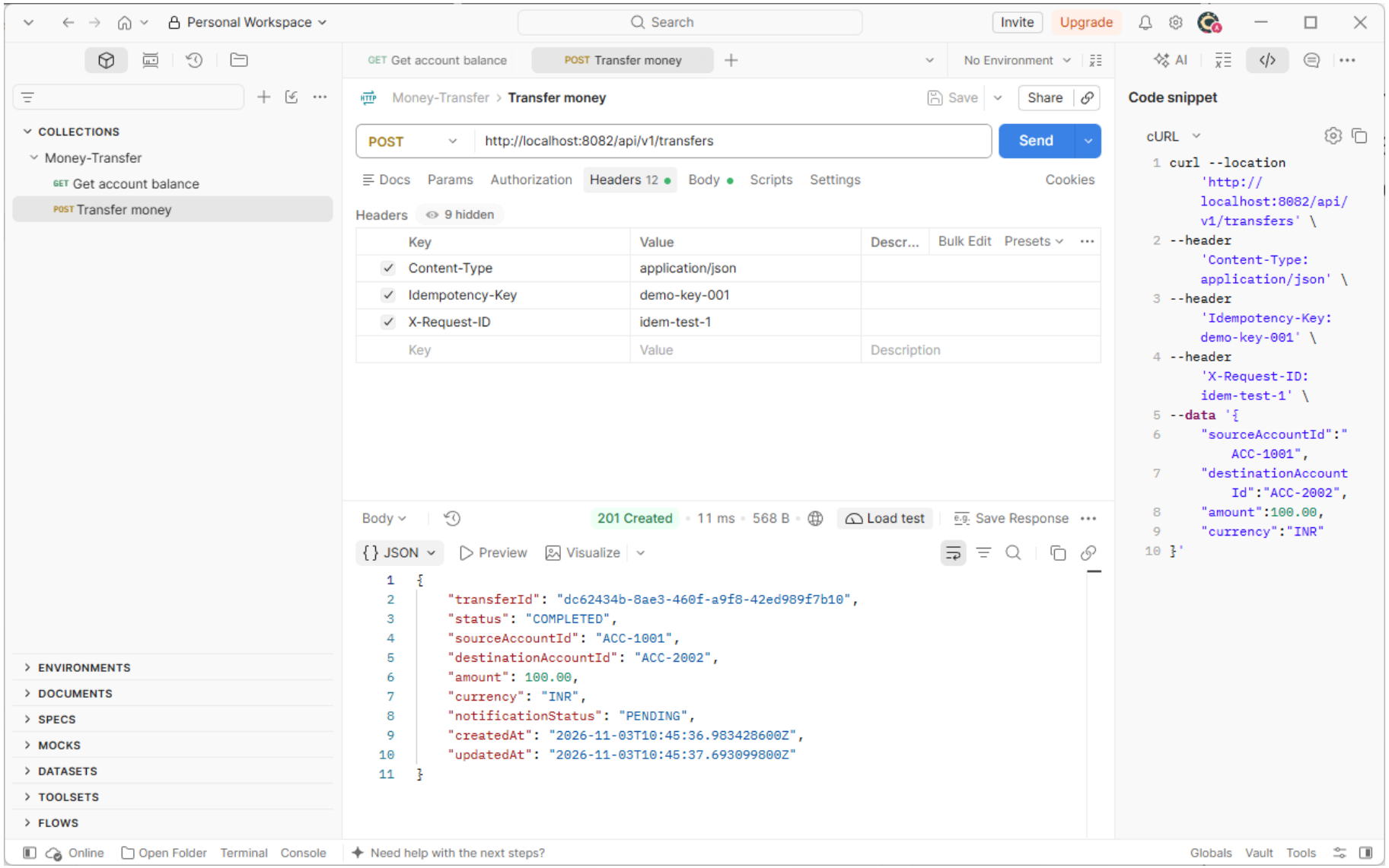
Checking account balance of account ACC-1002



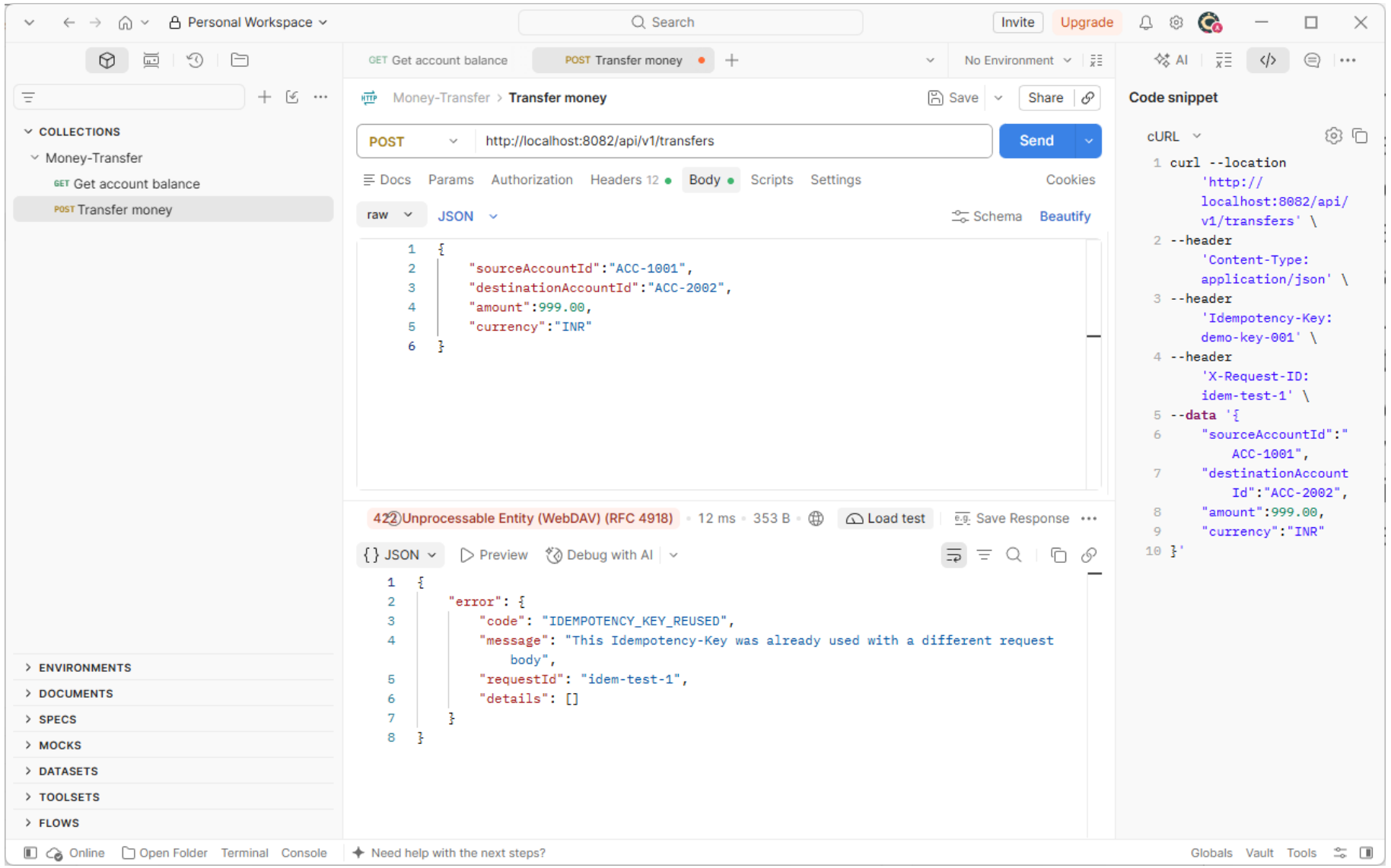
The transfers endpoint of transfer service, in the headers we pass an idempotency key and an Request ID



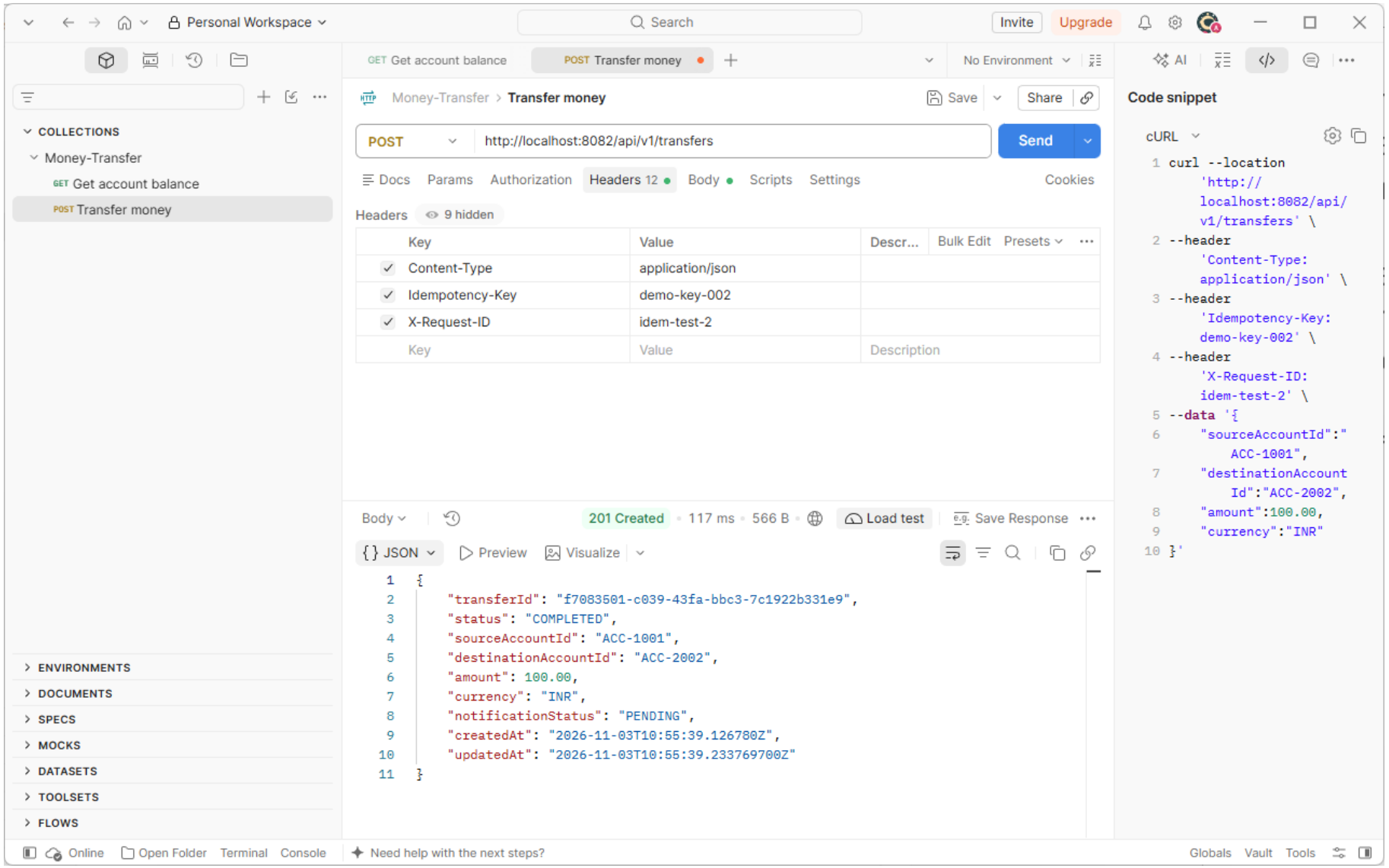
Calling the transfers endpoint of transfer service, in the body we pass source account id, destination account id, amount and currency. Here we are making an transfer of 100 INR from ACC-1001 to ACC-2002. In response we see the status as completed and the notification is yet to be sent.



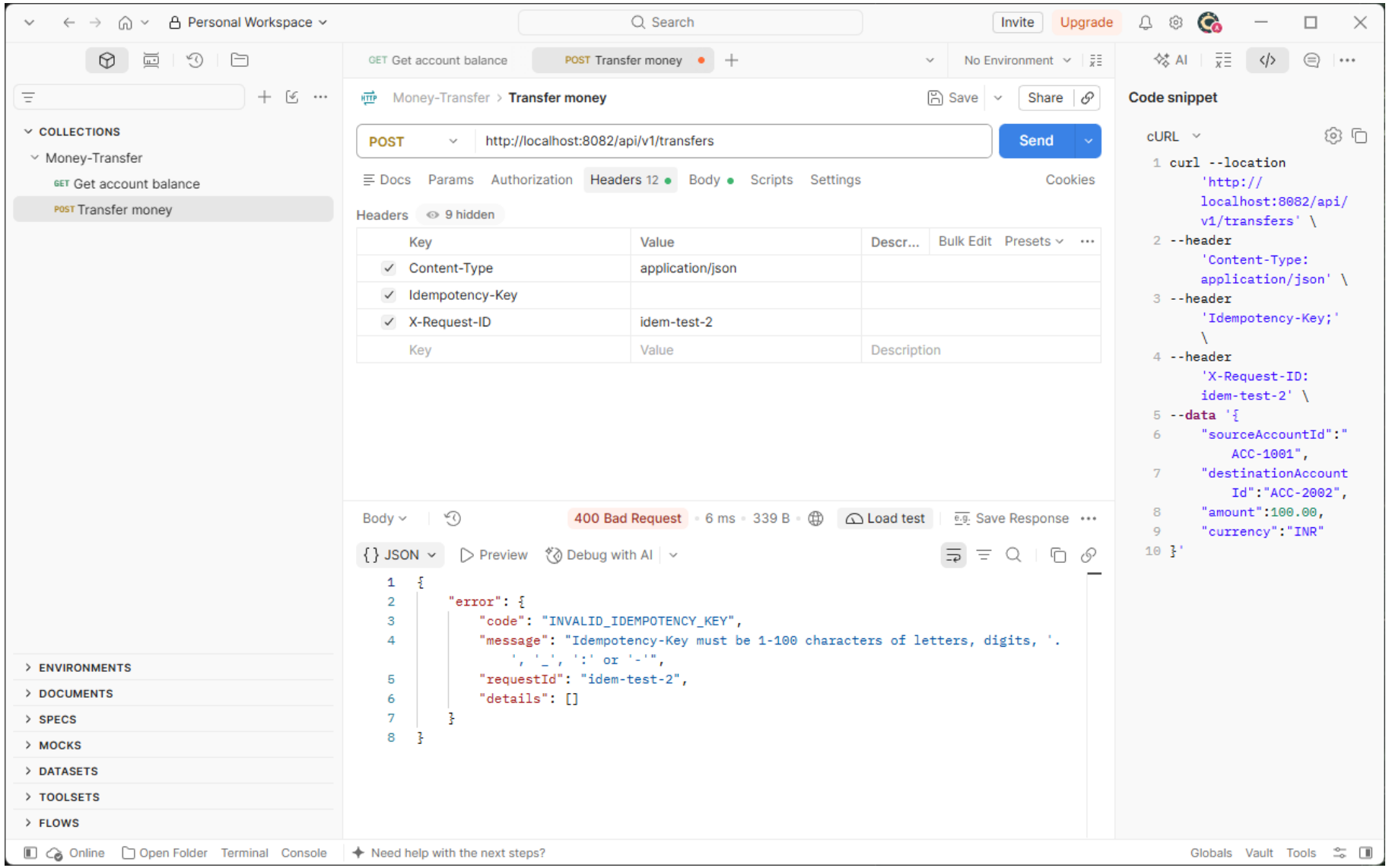
We called the same API again and you see the same response as the idempotency key is the same, we get the same output and application did not create a new transfer which we can verify using the transfer id in response it is the same.



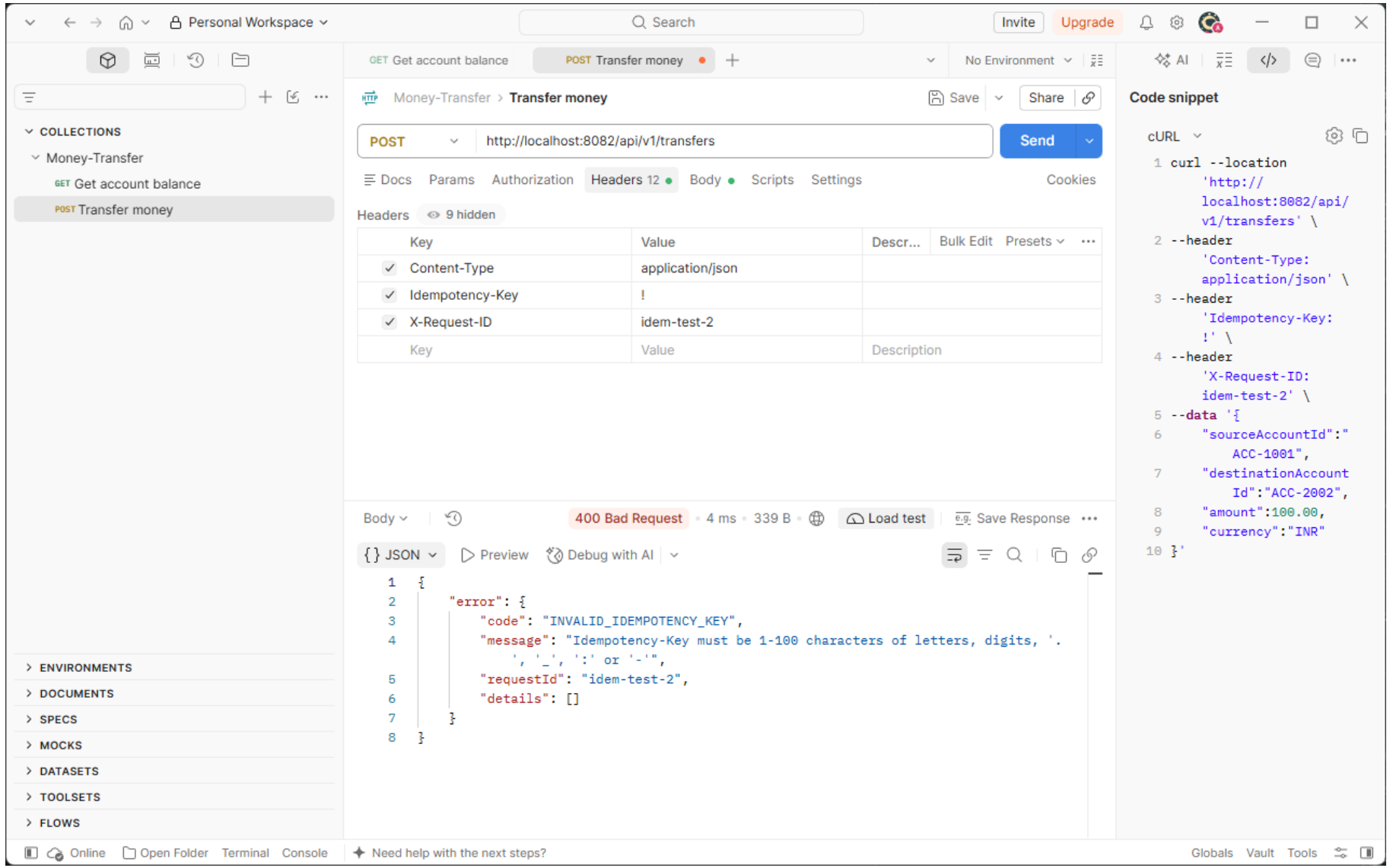
We called the same API again using the same idempotency key but a different amount, the response is an error stating the idempotency key is reused and there was no transfer made.



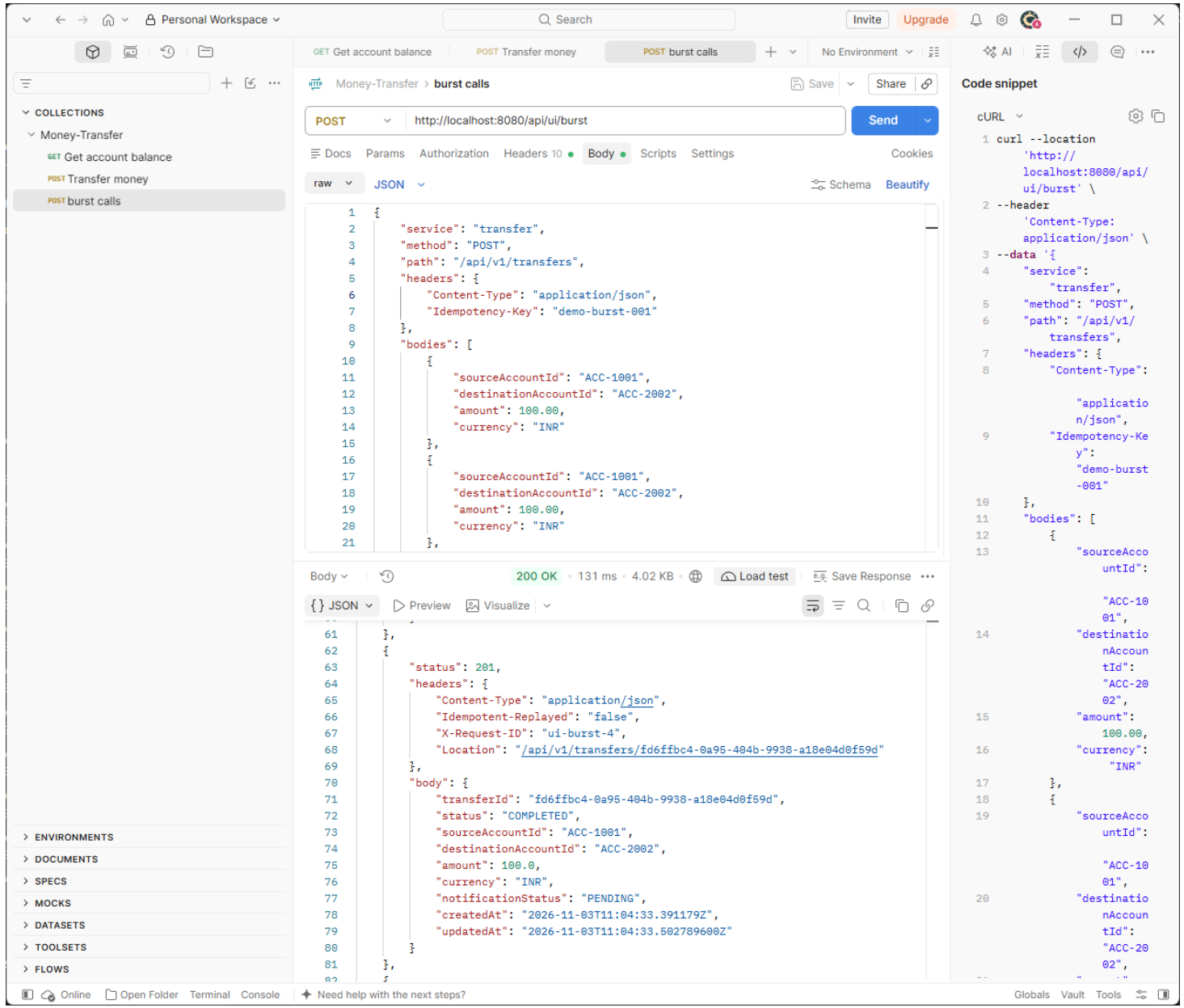
We called the same API again using a different idempotency key but the same amount, the response is details of a successful transfer as the service checks for a unique idempotency key and does not care about if the amount is same or not



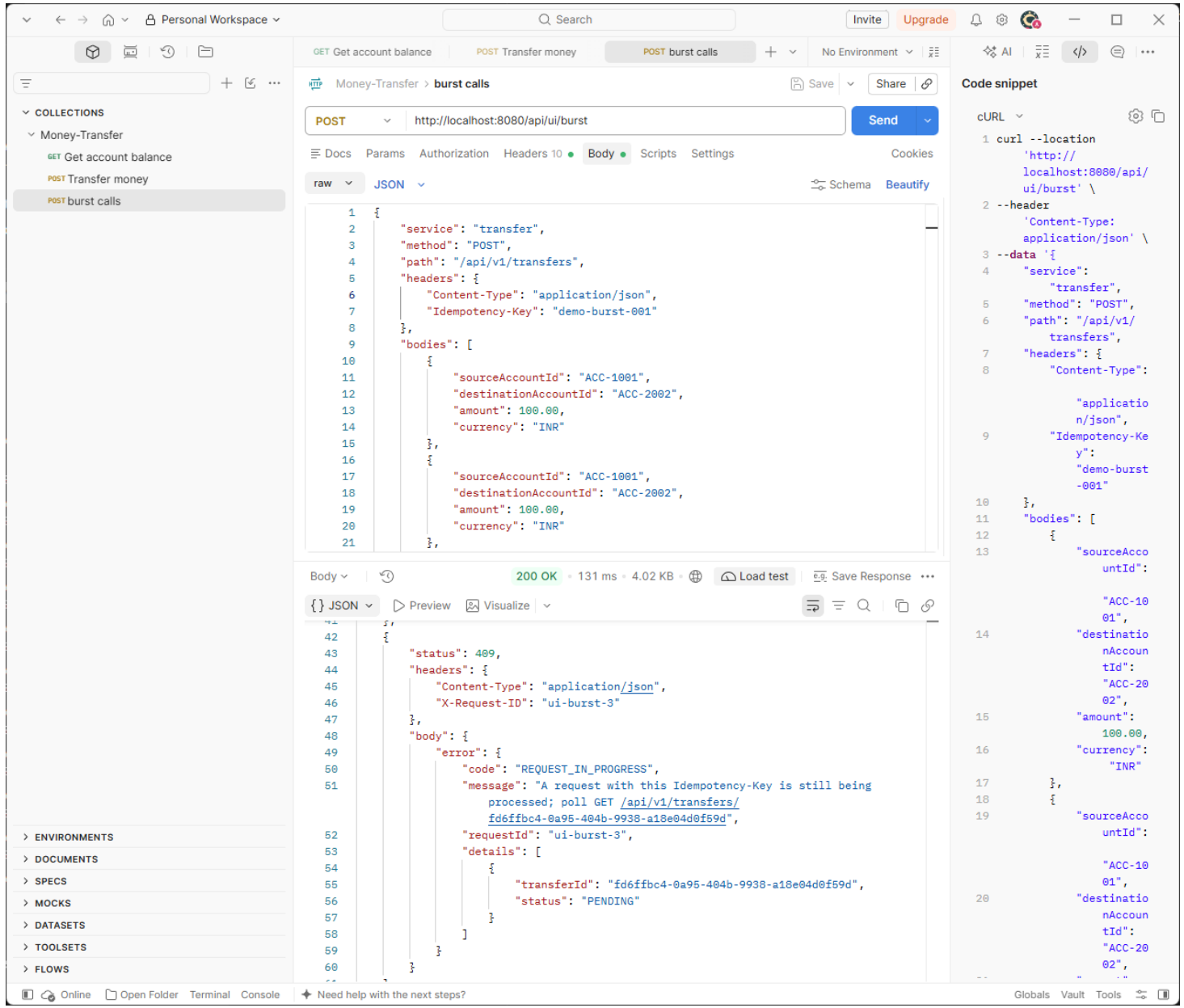
We call the transfers api but this time without an idempotency id, the response is an 400 as its an unsafe call and each request must carry a key.



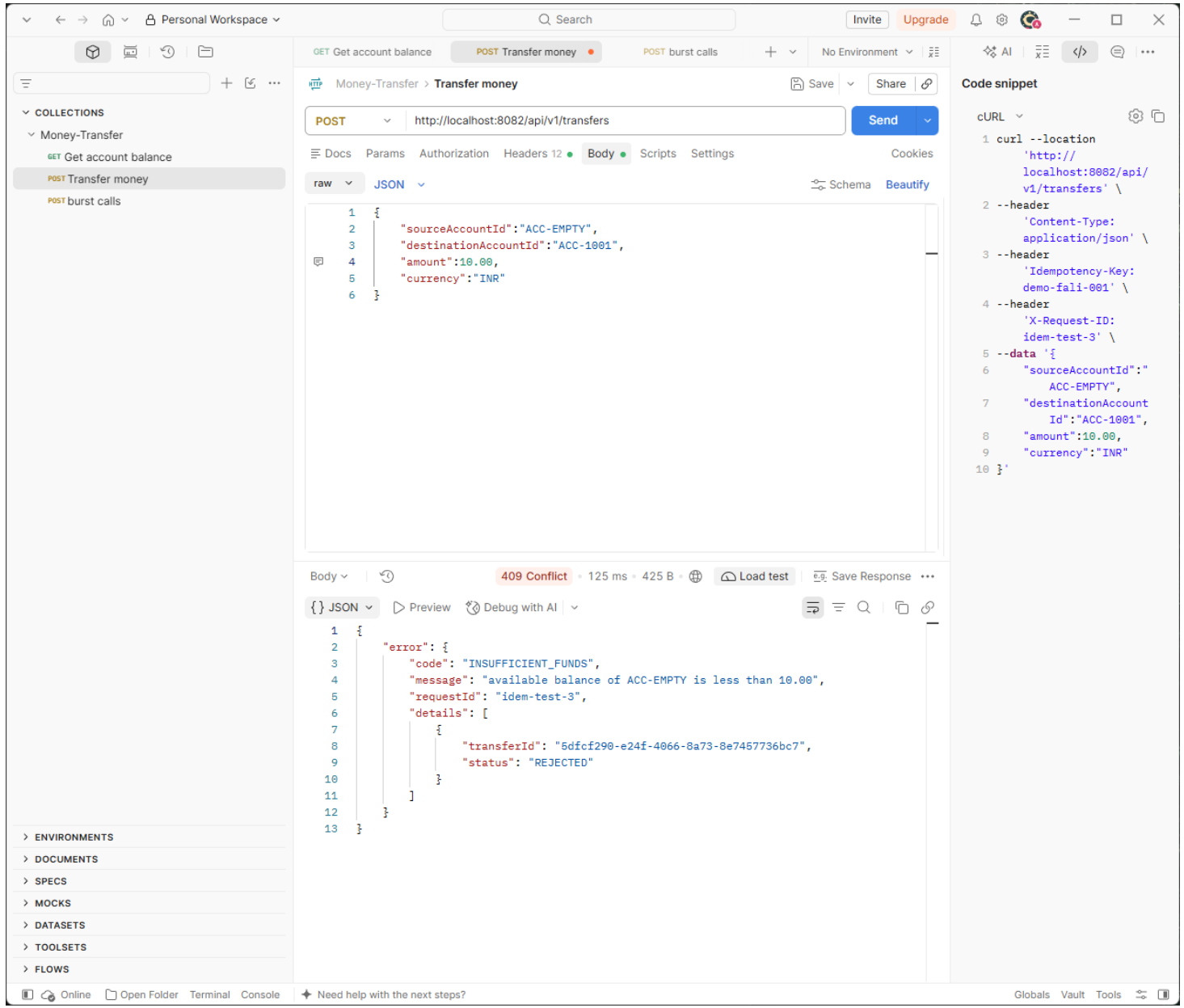
We call the transfers api and this time we sent an invalid idempotency key so it did not pass the idempotency key validation check and returned 400.
Keys must be 1-100 chars from letters, digits, ., _, :, -



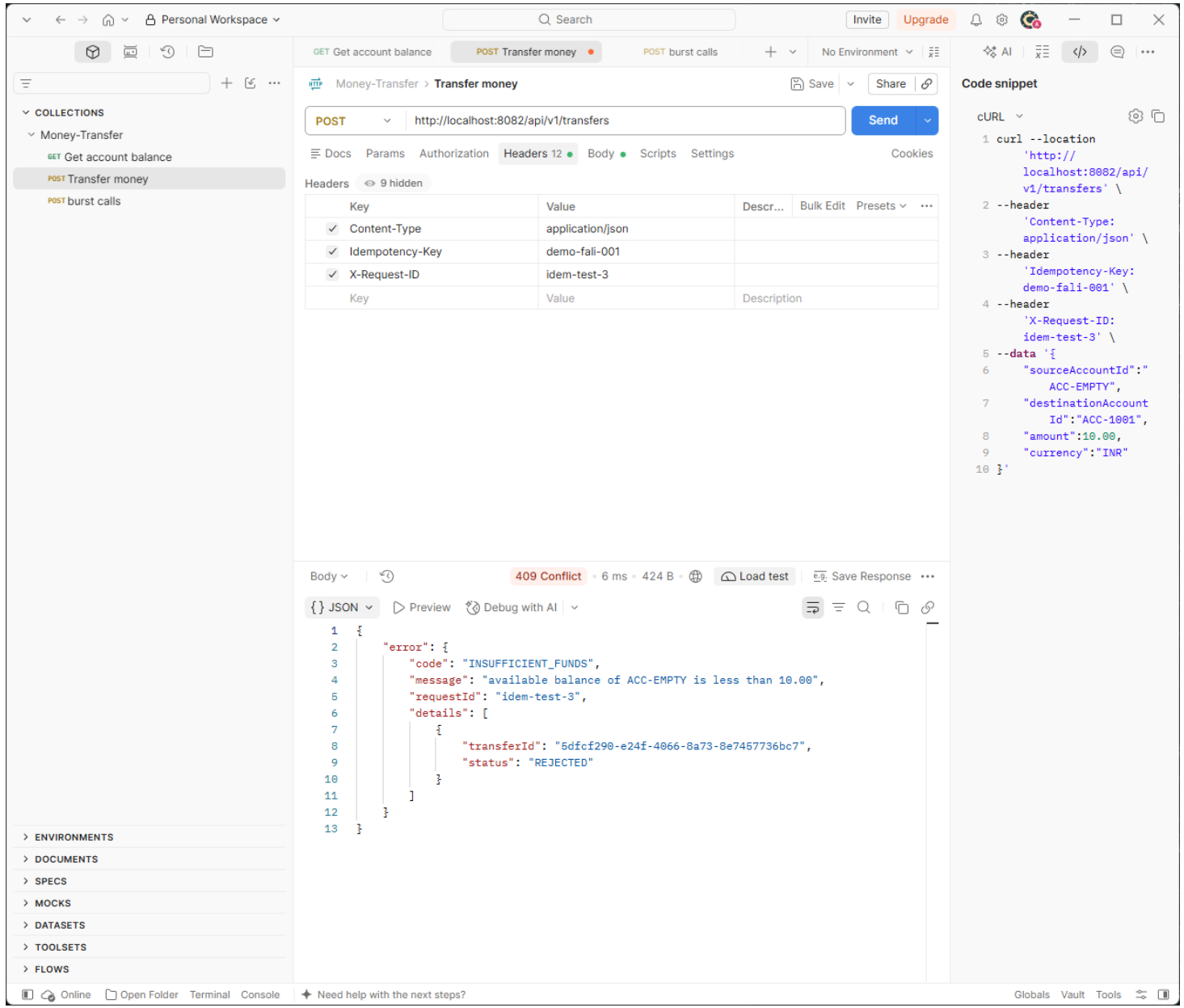
We will fire 10 identical requests at once, for this I have created a separate endpoint in a temporary service which I used for testing as postman cant fire burst requests.
In the respone you will see out of 10 identical requests one will be completed.



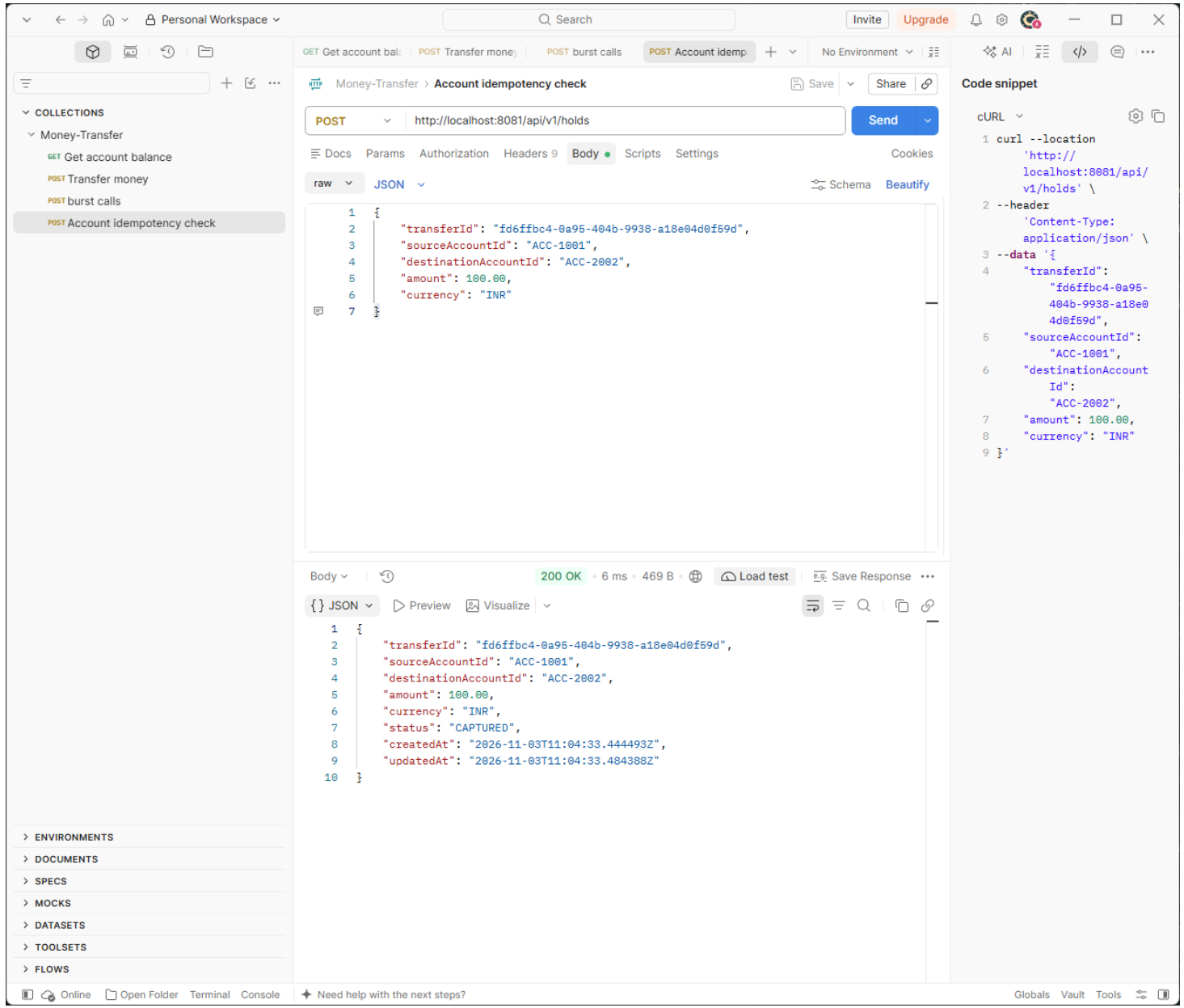
In the response for other identical rtequests except the one completed we can see that they give 409 and rightly point that the request with this idempotency key is processing and it also sends us the transfer ID generated for that request.



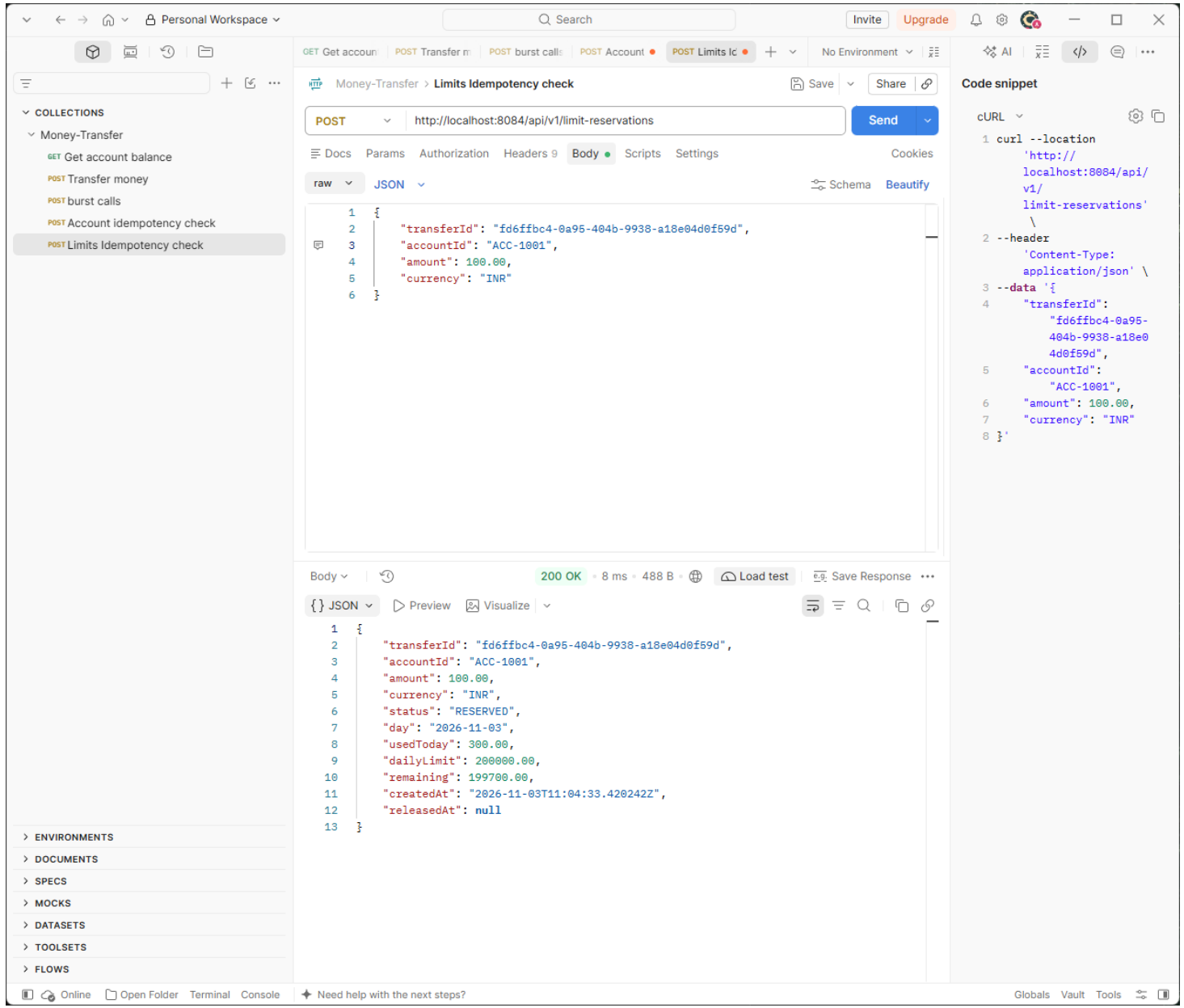
We try to make a transfer from an empty account, the response is valid insufficient funds. We also get a transfer ID in response.



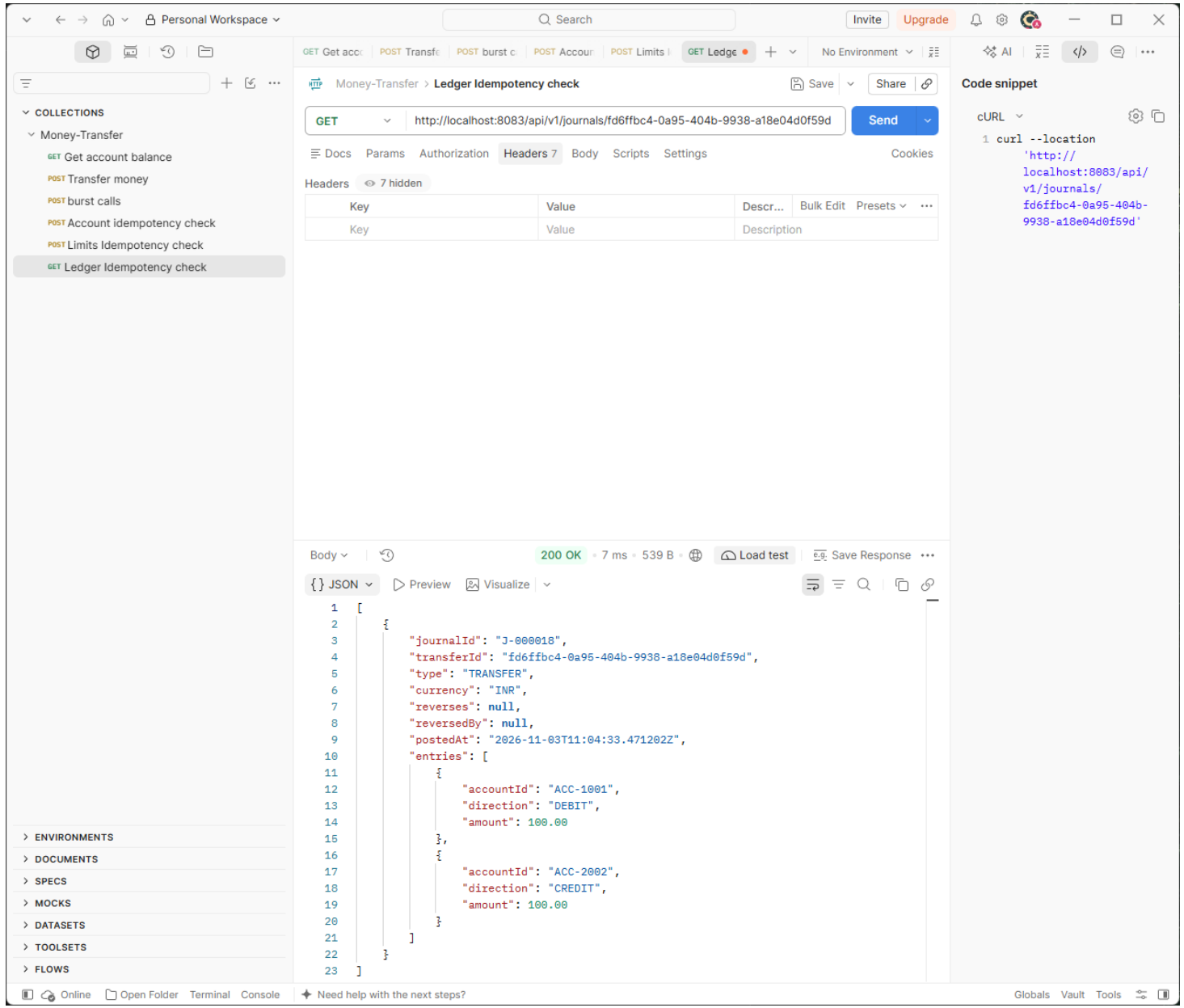
Ran the same identical request again and you can see it fails as insufficient funds but we also see that the transfer Id is still the same.
Even the failed transactions are stored and replayed and the transaction is not run again.



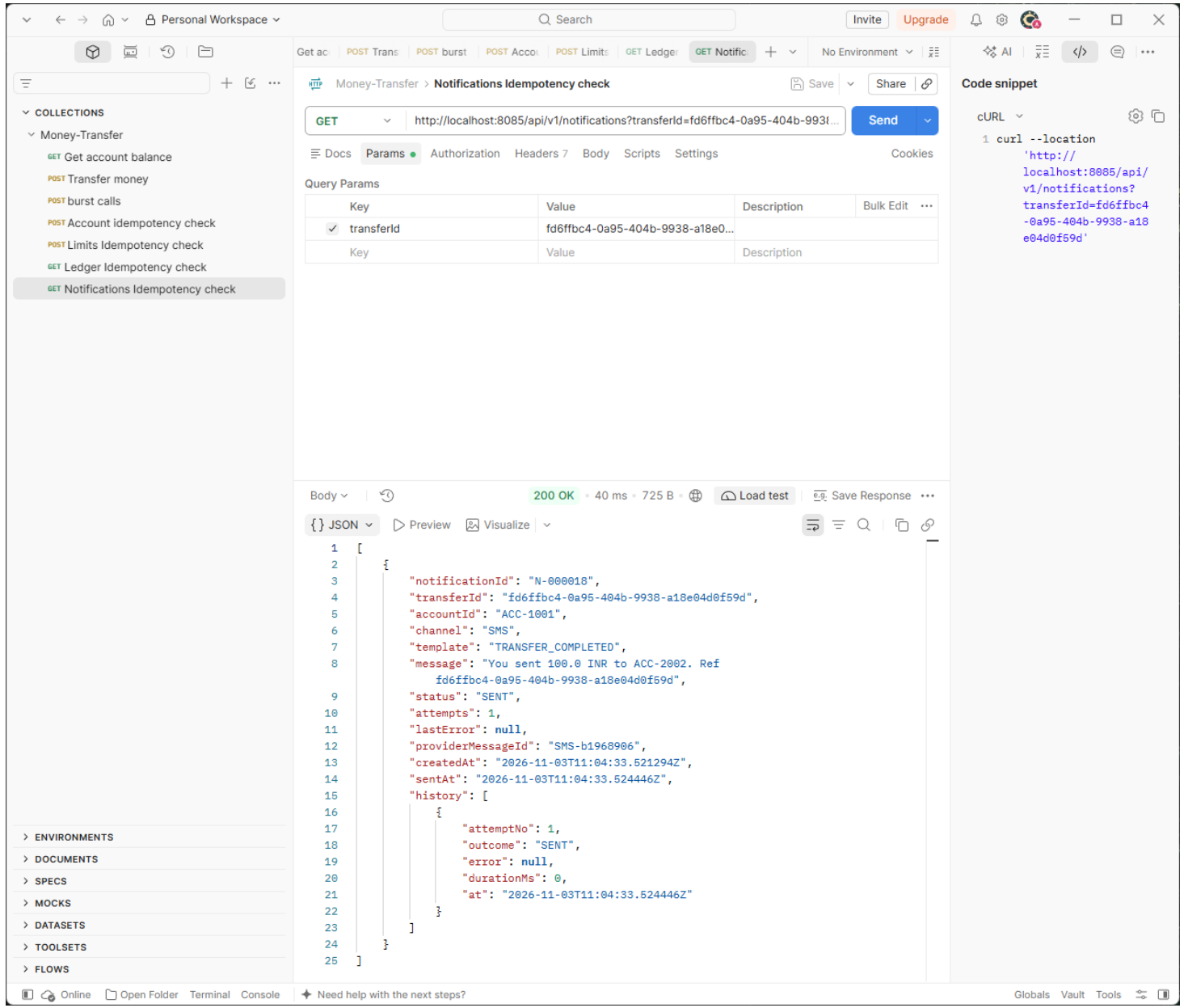
To check idempotency internally from the orchestrator to the layer 2 services.
We enter the same transfer id which is already completed.
We will see that it does no create a new hold instead return timestamp of the original transaction.



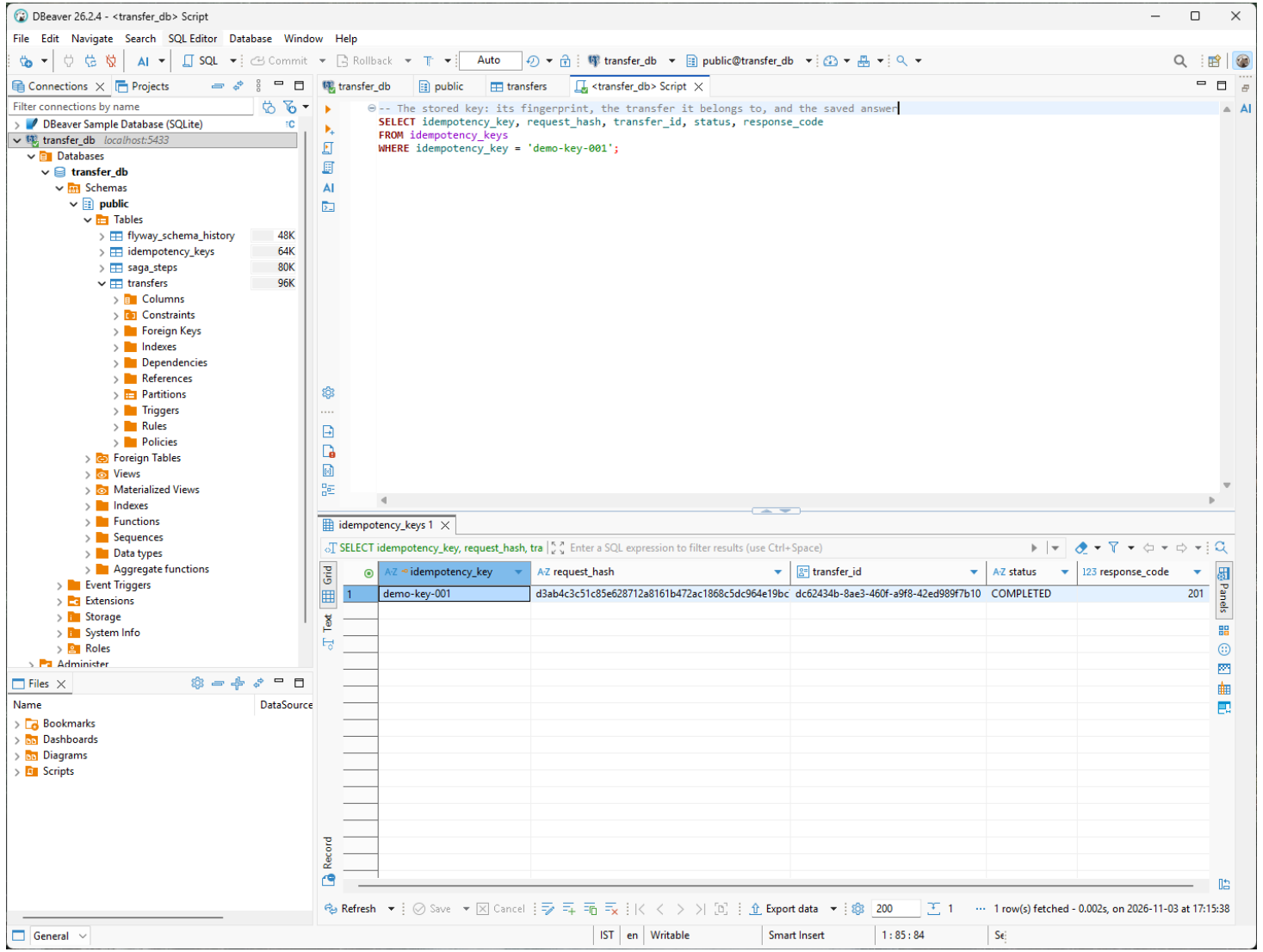
To check idempotency internally from the orchestrator to the limits service.
We enter the same transfer id which is already completed.
We will see that if we pass the same transfer id the remaining limit wont get changed.



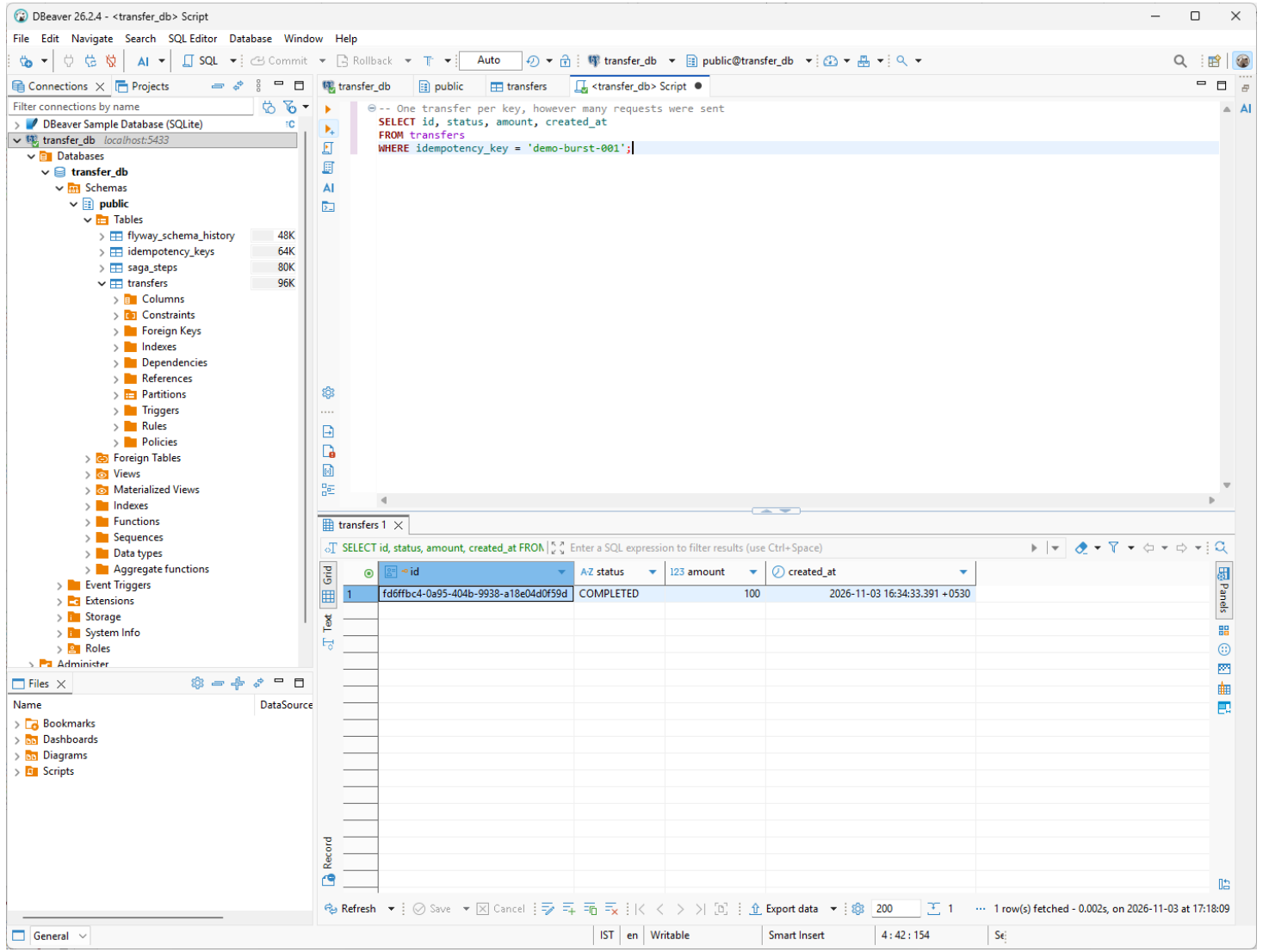
To check idempotency internally from the orchestrator to the ledger service.
We pass the same transfer id as param and there is a single journal entry.



To check idempotency internally from the orchestrator to the notification service. We pass the same transfer ID as param and we see only 1 notification entry however if there are multiple attempts to send notification it will reflect in the response.



We check now directly in the DB we find that for a idempotency key there is a request fingerprint, the related transfer id and the saved response.



For the idempotency key demo-butst-001 we fired 10 requests with the same idempotency key but only 1 succeeded and that is in the record rest did not create any new record.